

Katherine Langille

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SKILLS

Languages: Python, C++, SQL, JavaScript, HTML/CSS

ML & Deep Learning: PyTorch, Scikit-learn; transformers & attention, diffusion models (DDPM), model evaluation, interpretability & monitoring, responsible AI

Data & Analytics: Pandas, NumPy, Matplotlib, Jupyter, Power BI; data pipelines, process mining, dashboards, visualization

Tools & Systems: Git, Linux, Arduino; signal processing, embedded & real-time inference, hardware-software integration

EXPERIENCE

Machine Learning Intern

Jun. 2026 – Present

Signal 1

Toronto, ON

- Contribute to system for **governing and monitoring** machine learning and AI models in healthcare
- Build and test Python tooling to support **model evaluation and performance monitoring**
- Support validation of AI models to strengthen reliability and **responsible-AI standards**

Bioinformatics Software Research Student

Sep. 2025 – Dec. 2025

The Hospital for Sick Children (SickKids) — Centre for Applied Genomics

Toronto, ON

- Designed and implemented an **end-to-end machine learning pipeline** over demographic, clinical, and imaging (MRI) data to predict developmental outcomes
- Built, evaluated, and iterated on models with a focus on interpretability, robustness, and reproducibility
- Contributed to a **transformer-based protein language model** for variant-effect prediction, analyzing attention mechanisms to improve interpretability

Data Science and Process Mining Analyst

Oct. 2024 – Mar. 2025

Software Solutions, Transport Canada

Ottawa, ON

- Led two **process mining** projects, analyzing large-scale process logs in Python to identify workflow inefficiencies and deliver data-driven recommendations
- Designed a computationally efficient and secure **ML classification pipeline** to reduce administrative burden
- Built process conformance models and dashboards to translate insights into actions for cross-functional stakeholders

PROJECTS

Predictive Modeling of Human Gait | *Neuromechanics & Assistive Robotics Lab — Dr. Arami*

Winter 2026

- Worked on building a **multi-dataset pipeline** to harmonize heterogeneous human-gait datasets into a canonical representation via spatial mapping and temporal normalization
- Also worked on developing a deep probabilistic model (**denoising diffusion, DDPM**) to predict full gait cycles from past kinematics

EMG-Controlled Robotic Hand | [YouTube](#)

Summer 2025

- Built a low-cost system enabling **real-time grasp control** from forearm EMG signals in a 3-person team
- Designed and implemented **embedded signal-processing** and control pipelines to amplify, filter, and digitize signals for reliable real-time inference

Healthcare Systems Intervention Framework | *Innovation North Systems Innovation Challenge*

2026

- Designed a **systems-level intervention framework** linking housing affordability, family dispersal, informal care-network loss, and downstream hospital demand
- Synthesized cross-sector drivers into intervention directions around modular housing, mixed-community living, and preventative care capacity; awarded **“Best Story Told”** among 245 students from 25 universities

EDUCATION

University of Waterloo

Waterloo, ON

BASc. Biomedical Engineering, Honours, Co-Operative Program

Sep. 2023 – May 2028

- GPA: 3.95 (94.36%)**, First in Class 2B Term; Fairfax Financial Academic All-Canadian (Top-4 GPA among Student-Athletes)
- Competitions:** 1st Place, Bio-Engineering — Ontario Engineering Competition 2026; “Best Story Told” (\$5,000) — Innovation North Systems Innovation Challenge 2026 (245 students, 25 universities)
- Awards:** Dean’s Honours List, First in Class Engineering Scholarship, Ontario Professional Engineers Foundation Scholarship, President’s Scholarship of Distinction, Alumni @ Microsoft Scholarship in Eng., Warriors Women’s Excellence Award
- Engineering Society Rep; Varsity Squash; Campus Response Team Responder & Assistant Trainer